

The Magnetic Effect of Electric Current

A current-carrying wire deflects a compass needle — the discovery that electricity and magnetism are linked.

Current makes a magnetic field

In Activity 4.1, a magnetic compass is placed under a wire. When the switch is **ON** (current flows), the compass needle **deflects**. When the switch is **OFF**, the needle returns to its original direction.

The big result: a current-carrying wire produces a **magnetic field** around it. This is the **magnetic effect of electric current**. The field disappears when the current stops.

Key words

- **Magnetic field** — the region around a magnet or current-carrying wire where its magnetic effect (e.g. a compass deflection) can be felt.
- The compass needle is itself a **tiny magnet**, so it responds to the wire's magnetic field.

Who discovered it?

Hans Christian Oersted (Denmark, **1820**) noticed a compass needle deflect whenever a nearby circuit was switched on or off. He showed electricity and magnetism are **linked** — one of the great discoveries of science.

Uses: the magnetic effect runs electromagnets, electric bells, motors, fans and loudspeakers.

Where students lose marks

Trap 1 — No current, no field. The magnetic field exists only while current flows; it vanishes when the current stops.

Trap 2 — It is current, not the wire, that matters. A plain wire with no current does not deflect the compass.

Trap 3 — Oersted, 1820. Remember the name and that the discovery is the *link* between electricity and magnetism.

Direction: reversing the current direction reverses the direction of the needle's deflection.

Model answers — write them like this

Example A. Why does a compass needle deflect when current flows in a nearby wire?

Step 1: The current produces a magnetic field around the wire.

Step 2: The compass needle is a tiny magnet, so the field turns it.

∴ The current's magnetic field deflects the needle.

Example B. What happens to the needle when the current is switched off?

Step 1: No current → no magnetic field around the wire.

∴ The needle returns to its original (north–south) direction.

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